北京大學
PEKING UNIVERSITY

Thermodynamic Integration and Its Application to Disordered Materials

Group Meeting – March 13, 2026

Presenter: Chon-Hei Lo

Viewing high entropy alloys through glasses: Linkages between solid solution and glass phases in multicomponent alloys

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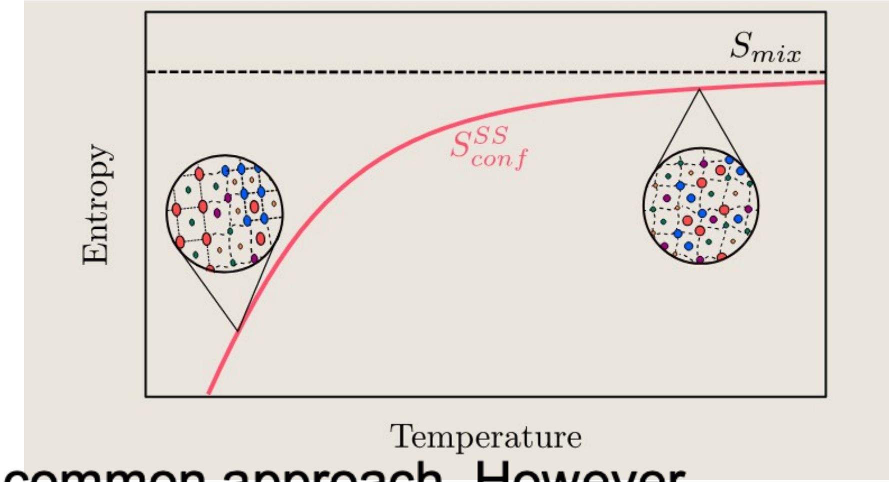
Concentrated Solid Solutions (CSSs)

CSSs (**高浓度固溶体**) are demonstrated extraordinary mechanical properties and radiation. One famous CSS is High Entropy Alloy (HEA).

1. Hardness (1GPa ~ 10GPa)
2. Corrosion resistance (Stainless Steel)
3. Oxidation resistance (up to 1400K)

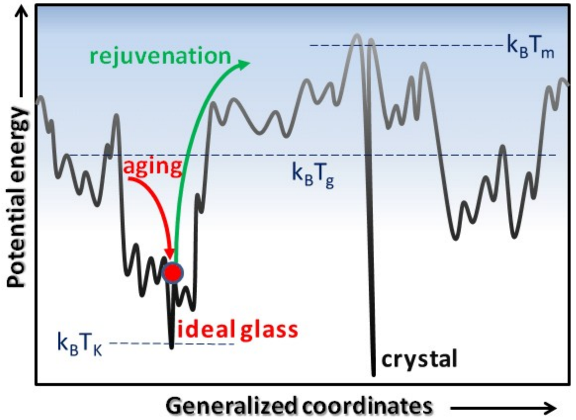


To design a CSSs, maximizing configuration entropy S_{conf} is a common approach. However, **nonequiatomic** (非等原子比) is often better. ($Zr_{44}Ti_{11}Ni_{10}Cu_{10}Be_{25}$, $Zr_{55}Cu_{30}Ni_{5}Al_{10}$,…)

Reason: S_{mix} is the limit of no interaction S_{conf}.

Conflict: $S_{conf} < S_{mix} \Rightarrow \max\{S_{\mathrm{conf}}\} \neq \max\{S_{\mathrm{mix}}\}$.

$$S_{\mathrm{conf}}(E_0) = k_B \ln\left(\int_{E<E_0} \mathrm{d}\rho(E)\right) \quad S_{\mathrm{mix}} = k_B \sum_i -c_i \ln c_i$$

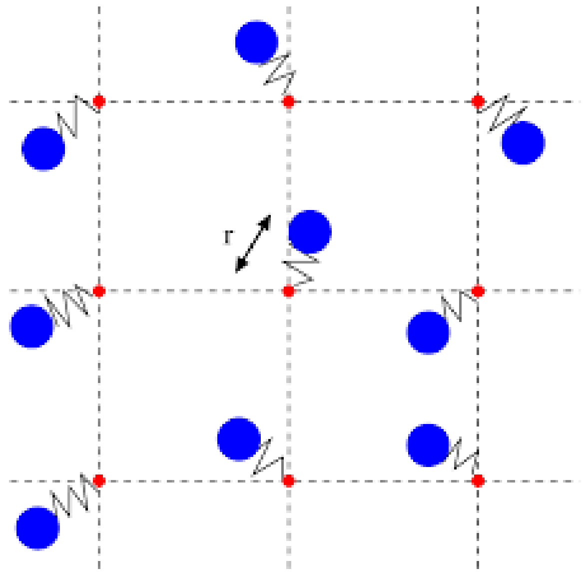


How can we find free energy $F(T_0)$

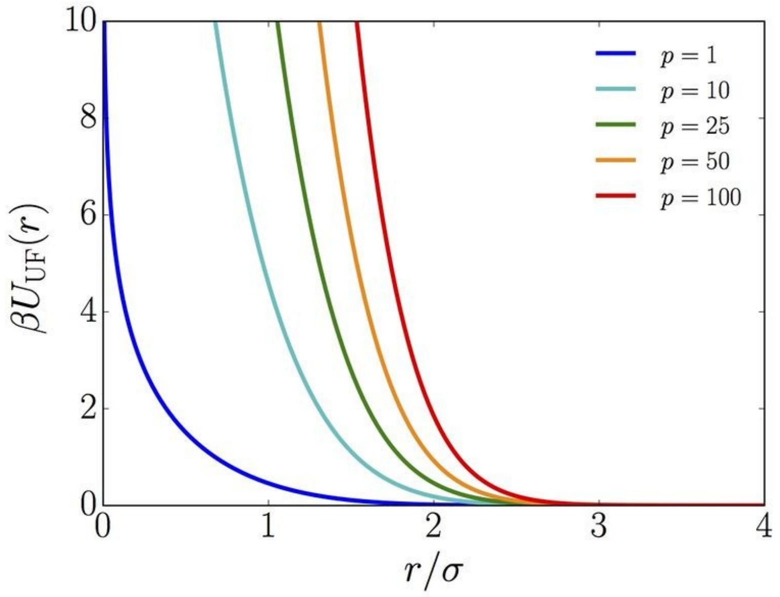
To find $F(T_0)$, we have several approach using **thermodynamic integration(热力学积分)** depending on the simulation setting:

1. Einstein Crystal - No interaction. Atoms are pinned by harmonic oscillators.
2. Uhlenbeck-Ford model (Gaussian gas):

$$U_{\mathrm{UFM}} = -k_B T \ln\left(1 - e^{-r^2/\sigma^2}\right)$$



Simulate the overall system using $U(\lambda) = (1-\lambda)U_{\mathrm{Model}} + \lambda U_0$. The overall free energy: $G(T) = \underbrace{G_0(\mathrm{T})}_{\text{model free energy}} + \int_0^1 \frac{\partial U(\lambda)}{\partial \lambda} \mathrm{d}\lambda$



Free energy depending on temperature $F(T)$

Reversible scaling (可逆缩放法) is a technique to calculate entropy(S) and free energy(G) on-the-fly during the simulation.

Let $\zeta(t)$ is a time-depending parameter. The system is evolving through:

$$\hat{U}(t) = \zeta(t)U$$

In such system, the temperature is equivalent to $T = \frac{T_0}{\zeta(t)}$ with potential U.

We don't change the temperature.

The free energy is:

$$G(T) = \frac{1}{\zeta(t)}\left(G(T_0) + \frac{3}{2}Nk_BT_0 \ln \zeta(t) + W(t)\right),$$

where irreversible work $W(t) = \int_0^t U\big(R(\tau)\big)\frac{\mathrm{d}\zeta}{\mathrm{d}\tau}\mathrm{d}\tau$.

We can find $S_{conf}(T)$ after all these process

Free energy depending on temperature $F(T)$

For irreversible work $W(t) = \int_0^t U(R(\tau)) \frac{\mathrm{d}\zeta}{\mathrm{d}\tau} \mathrm{d}\tau$. There are two types of errors:

1. statistical error – depends on initial configuration $R(t)$.
2. systematic error – dissipative entropy(耗散熵).

Assuming the system is in **Linear Response Region**, the dissipative can be cancelled through averaging forward and backward process:

$$\mathcal{W}_{T_0 \to T} = \Delta E_{\text{diss}} + \mathcal{W}_{qs}$$

$$\mathcal{W}_{T \to T_0} = \Delta E_{\text{diss}} - \mathcal{W}_{qs}$$

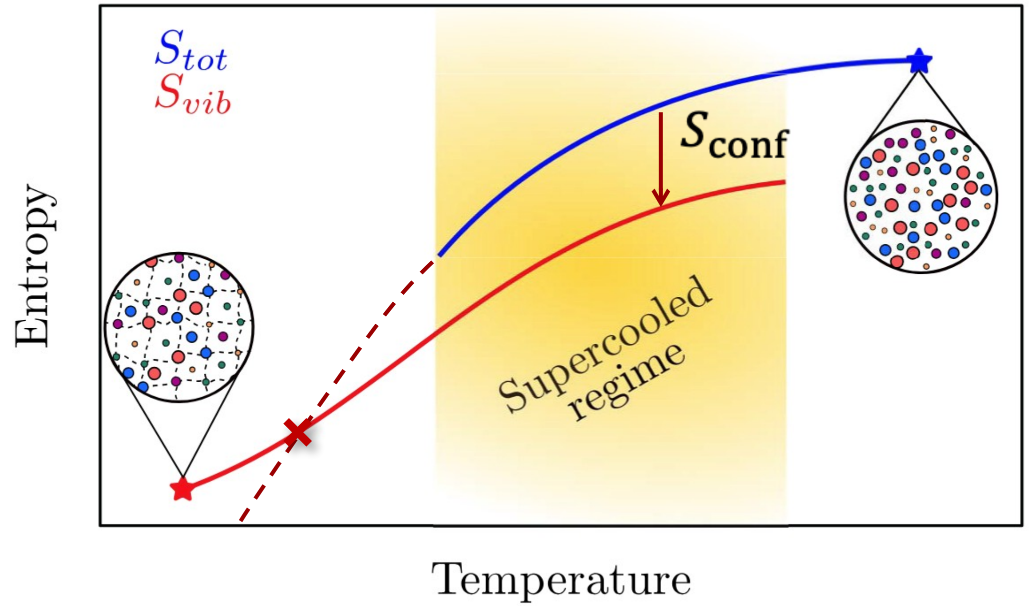
Configurational entropy $S_{\text{conf}}(T)$

Glass state

Einstein Crystal is calculating S_{vib}

UFM model is calculating S_{tot}

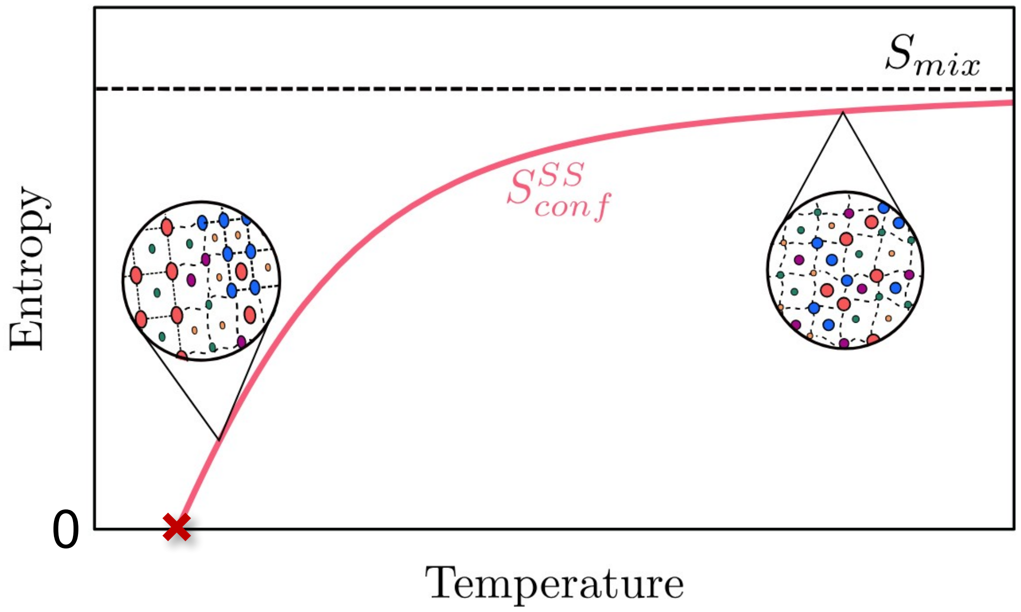
The difference of them is S_{conf}



Solid solution (Crystal)

The lattice is fixed. So, running swap MC on extremely high temperature still valid:

$$\lim_{T\to\infty} S_{\text{conf}}(T) \to S_{\text{mix}}$$



Example: $Co_{20}Cr_{20}Ni_{20}Fe_{20}Mn_{20}$ Cantor alloy

Second-nearest neighbor modified embedded atom method(2NN-MEAM).

Liquid region (Kubaschewski)

$$C_p = 3k_B + aT + \frac{b}{T^2}$$

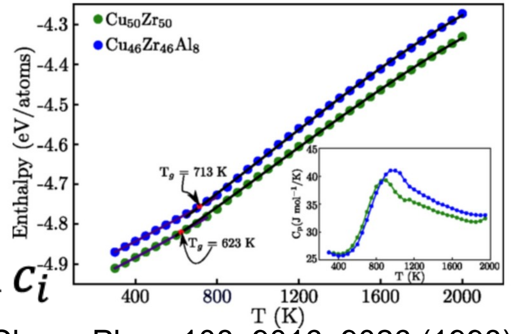
Solid region (Dulong-Petit)

$$C_p = 3k_B + cT + dT^2$$

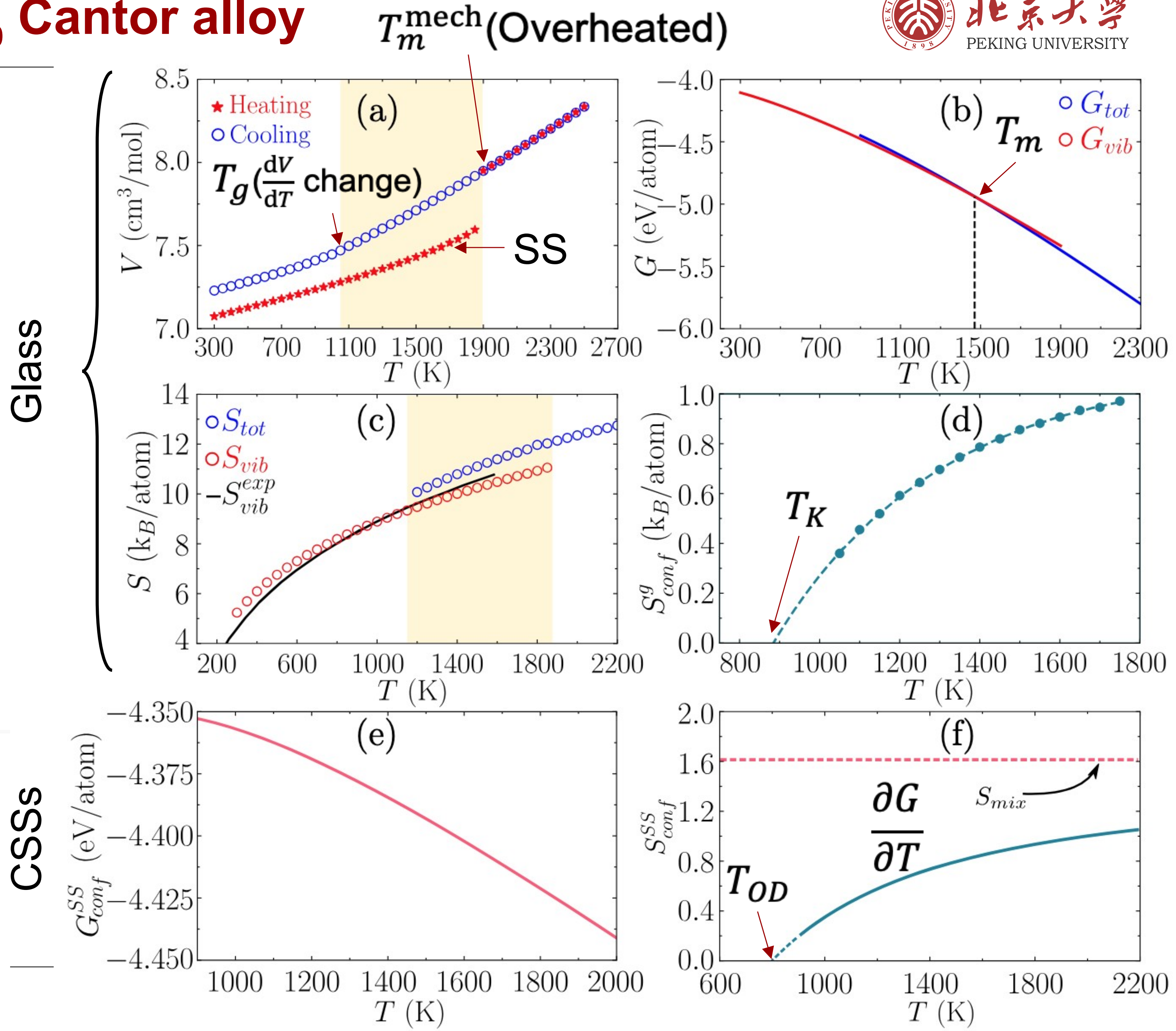
From heat compacity[1]:

$$TS_{\mathrm{conf}} = A(T - T_k) + B(T - T_k)^2$$

Example of T_g



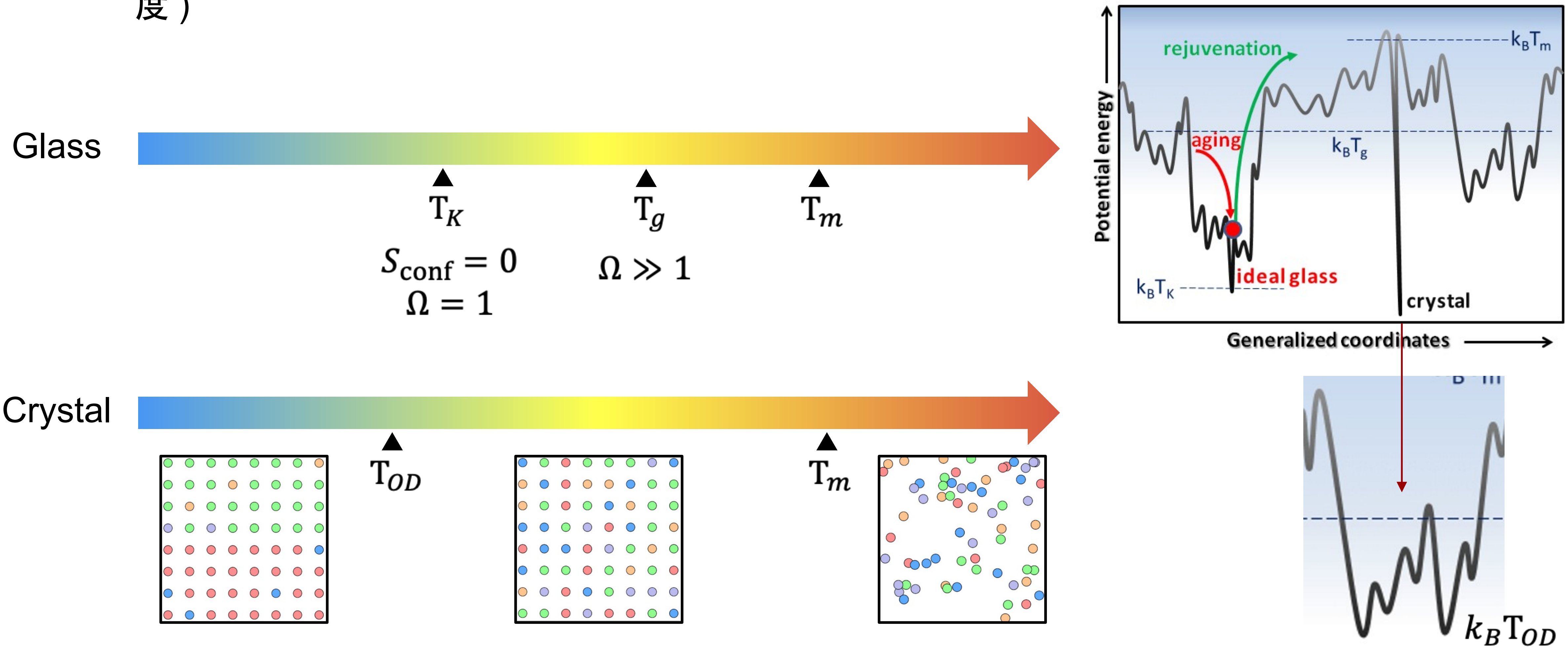
$$S_{\mathrm{mix}} = k_B \sum_i -c_i \ln c_i$$



[1] R. Richert, C. A. Angell, J. Chem. Phys. 108, 9016–9026 (1998)

Kauzmann Temp. and Order-Disorder Temp.

（考兹曼温度）



Thermodynamic properties of High Entropy Alloy and Glass.

1. $T_{OD} < T_K < T_g < T_m$ and $T_m \sim 1.5 T_g$.

2. S_{conf}^{g} change (%) doesn't seem directly linked to the element count → $S_{\mathrm{mix}} \neq$ Glass Forming Ability
(Making metallic glass is very hard!)

3. S_{conf}^{SS} change is related to the element count.

Alloy	T_m^{expt} (K)	T_m (K)	T_g (K)	T_K (K)	T_{OD} (K)
CoCrNiFeMn	1553	1470	1050	886	799
CoCrNiMn	1489	1465	950	768	605
CoCrNiFe	1695	1730	1150	949	681
CoNiFe	1724	1740	1050	831	770
FeNiMn	1473	1500	900	768	612
FeNiCr		1690	1150	1000	712
FeNi	1703	1670	1150	904	818

Stability of HEA – Short Range Order

T_{OD}

Define: $\alpha_{ij}^{(2)} = 1 - \frac{P_{ij}^{(2)}}{C_j}$, C_j is concentration(浓度).

No interaction: $P_{ij}^{(2)} = C_j \rightarrow \alpha_{ij}^{(2)} = 0$.

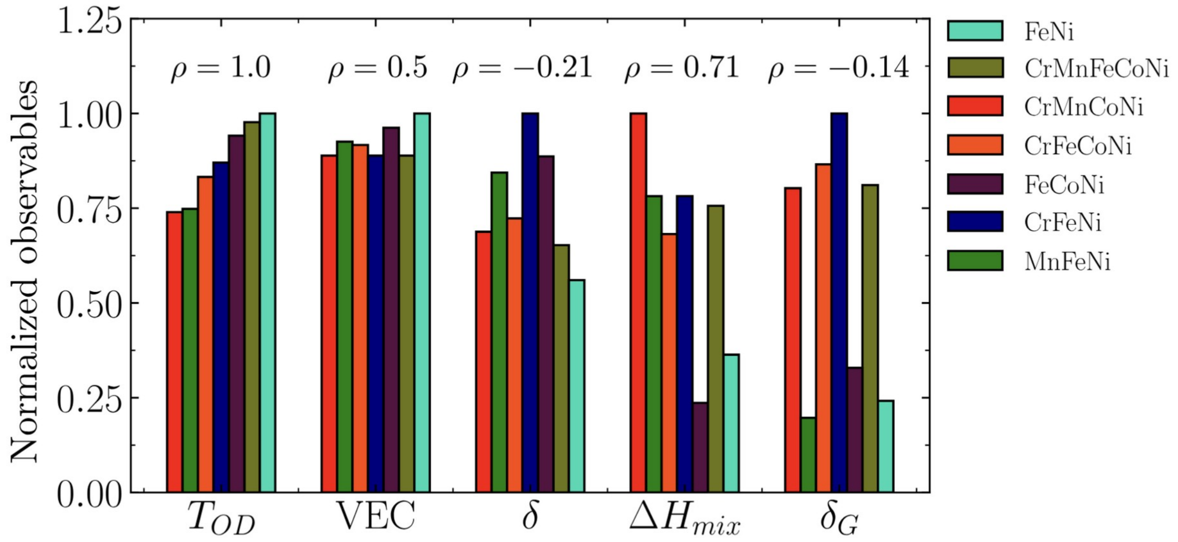
j condense at 2nd shell: $\alpha_{ij}^{(2)} < 0$.

VEC: valence electron concentration (价电子浓度)

δ: atomic size mismatch

ΔH: mixing enthalpy

δ_G: shear modulus mismatch



Fragilities (脆性) of HEA

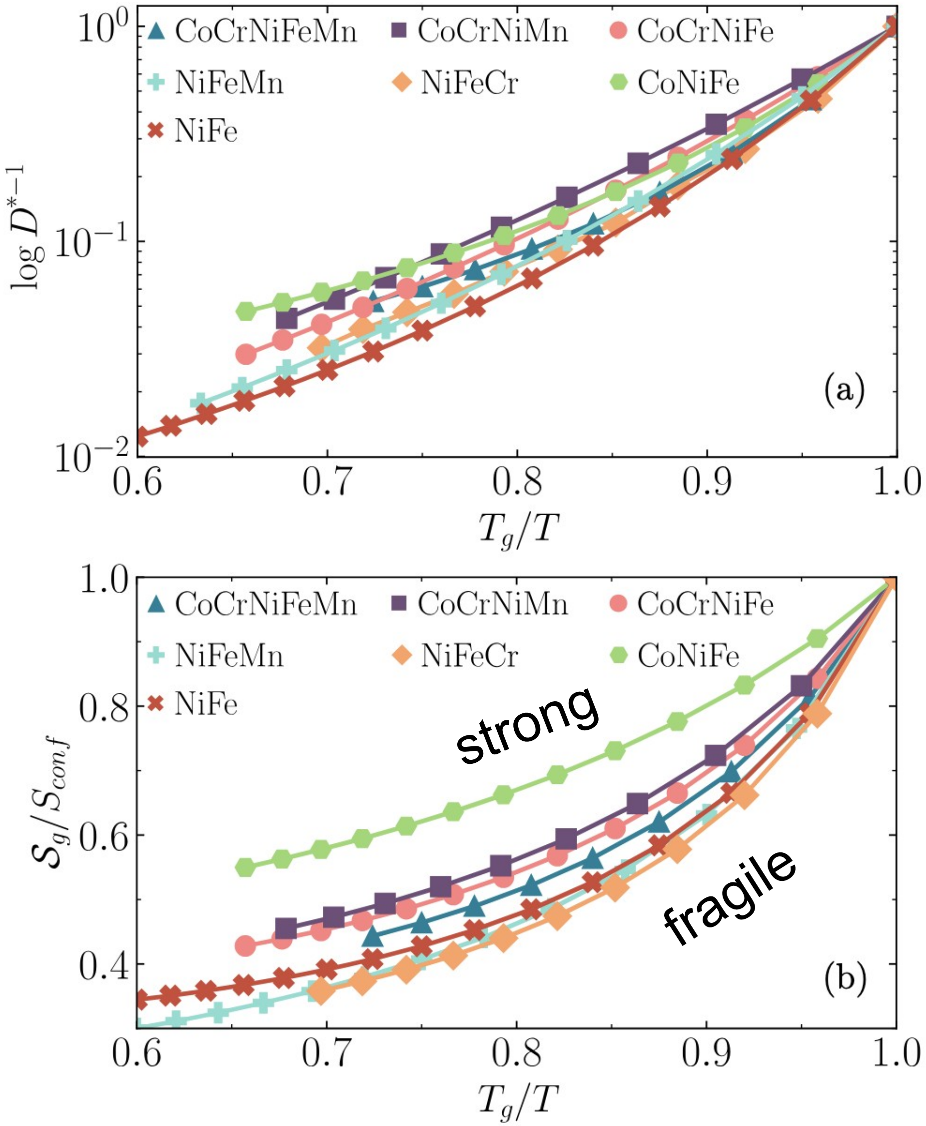
$$m_k = \left.\frac{\partial \log D^{-1}}{\partial\,(T_g/T)}\right|_{T=T_g},\quad m_t = \left.\frac{\partial (S_{\mathrm{conf}}^{-1})}{\partial\,(T_g/T)}\right|_{T=T_g}$$

kinetic fragility – the slowing in the diffusion

thermodynamic fragility – number of accessible state

$\frac{1}{D}$ ~ Viscosity (黏度)

From thermodynamics to kinetics!



Summary

1. 系统性使用了热力学积分去求一个无序体系（结构上的、还有晶格上原子）的自由能。并通过热力学理论去分析。

2. 外推了T_K的使用范围，提出了T_{OD}，相当于用玻璃角度理解晶体。

3. T_{OD}能够作为一个表征HEA相稳定性的参数。作为纯热力学的参数，有一定的动力学关联性。